

JUnit Testing Exercises

Exercise 1: Setting Up JUnit

Steps to setup project structures and default JUnit testing dependencies in JUnitDemo project.

Project: **JUnitDemo**

GroupId: **com.example**

Exercise 3: Assertions in JUnit

AssertionsTest.java

```
package com.example; import static org.junit.jupiter.api.Assertions.*; import
org.junit.jupiter.api.Test; public class AssertionsTest { @Test void passingTest() { int a = 10;
int b = 5; String name = "Sasank"; String empty = null; assertEquals(15, a + b); assertTrue(a >
b); assertFalse(a < b); assertNull(empty); assertNotNull(name); assertNotEquals(20, a + b);
assertSame(name, name); assertNotSame(name, new String("Sasank")); assertEquals(new int[]{1,
2, 3}, new int[]{1, 2, 3}); assertThrows(ArithmeticException.class, () -> { int x = 10 / 0; }); }
@Test void failingTest() { assertEquals(10, 2 + 3); assertTrue(5 < 3); assertFalse(5 > 3);
assertNull("Hello"); assertNotNull(null); } }
```

JUnit execution outputs with assertions:

JUnit Test Runner Results

Runs: 2/2 Errors: 0 Failures: 1

[X] AssertionsTest [Runner: JUnit Jupiter]

● passingTest() [0.015s]

■ failingTest() [0.002s] - org.opentest4j.AssertionFailedError

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures

CalculatorTest.java

```
package com.example; import static org.junit.jupiter.api.Assertions.assertEquals; import
org.junit.jupiter.api.AfterEach; import org.junit.jupiter.api.BeforeEach; import
org.junit.jupiter.api.Test; public class CalculatorTest { private Calculator calculator;
@BeforeEach void setUp() { calculator = new Calculator(); System.out.println("Setup completed"); }
@Test void testAddition() { int a = 10; int b = 5; int result = calculator.add(a, b);
assertEquals(15, result); } @AfterEach void tearDown() { calculator = null;
System.out.println("Cleanup completed"); } }
```

Calculator execution console logs:

Console Log Output

```
Setup completed
Running testAddition...
Cleanup completed
```

```
Setup completed
Cleanup completed
```

```
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.045 s - in com.example.CalculatorTest
```